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Systems, methods, and apparatus for mechanically removing liquid from material

Abstract

A system for dewatering a material comprising a slitter, wherein the slitter receives the material, separates the material into a plurality of clumps, and deposits the plurality of clumps of material substantially evenly on a conveyor belt. The conveyor belt is partially porous to allow water to pass through but preventing material from passing through. The conveyor belt is operable to convey the material from the slitter to a compression zone; the compression zone comprises at least one high pressure press. The compression plates engages the material positioned on the conveyor belt. At least one knife positioned proximate the at least one compression plate operable to remove material from the bottom surface of the at least one compression plate after a compression cycle; and at least one drain positioned under the conveyor belt to carry water removed from the material away from the conveyor belt.

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Background/Summary

(1) This application claims priority to provisional application Ser. No. 63/285,508 filed on Dec. 3, 2021 titled Systems and Methods for Mechanically Removing Liquid from Material, the entire content of which is incorporated herein by reference.

BACKGROUND OF INVENTION

(1) The disclosed inventions relate to novel systems, apparatuses, and methods to remove liquid from material. In most applications, the material comprises biosolids, e.g., the solid, semi-solid, or liquid residue generated during the biological wastewater treatment process. In other applications, the material comprises organic or inorganic material such as feed additives. The term “material” is used collectively to include biosolids and other organic or inorganic compositions.

(2) The disclosed inventions may operate independently to mechanically dewater material. Alternatively, the disclosed inventions may supplement existing systems or methods used to generate class A fertilizer from biosolids as described in U.S. Pat. Nos. 9,751,813 and 10,259,755 and pending application Ser. No. 17/220,994, all of which are owned by the instant applicant, and the entire disclosures of which are incorporated herein by reference.

SUMMARY OF INVENTION

(3) The disclosed inventions process material (e.g., sludge) that may have been dewatered with a belt press, centrifuge, screw press, or other type of dewatering device. The inventions mechanically compress the material (sludge) to remove water from the material that cannot be extracted through conventional means. Material such as sludge starts out as a liquid and can be dewatered by adding polymer. Adding polymer removes water from the material to increase the percent solids of the material. The dewatered material may then be fed into the double drum dryer to remove additional water to increase the percent solids to a concentration of about 90%.

(4) The disclosed inventions apply high pressure forces to the material. A belt transports material on a woven mesh belt to a compression zone. Hydraulically actuated plates exert high pressure forces on the material to press additional water out of the material. This process increases the solids percentage from 15-25% to about 30-45%. The reduction in water content reduces the volume of material. When used in conjunction with existing systems (e.g., a double drum dryer) the disclosed inventions materially increase the throughput (i.e., the end product) because less water remains in the material before being exposed to a thermal drying system.

(5) The disclosed inventions may also allow liquid sludge to be delivered directly into the system with or without first adding a polymer to the sludge. One of the existing challenges in the industry is to dewater material as much as possible mechanically. Mechanical dewatering includes any system or process that does not use heat or thermal energy; therefore, mechanical dewatering costs less than heat or thermal drying. In one embodiment, a polymer is added to the sludge, the sludge is then processed through a centrifuge, belt press, or screw press. In this embodiment, the cost is about $0.01-$0.02 per gallon to remove a gallon of water from of a material.

(6) Thermal drying costs about $0.10-$0.20 per gallon to remove a gallon of water. The disclosed inventions reduce the cost to about 10% of the cost of thermal systems. Plants, such as municipal wastewater treatment facilities, that have moderate to low disposal costs focus on mechanical

dewatering. Larger volume plants that have higher disposal costs and want to achieve a Class A fertilizer end product will use mechanical dewatering and thermal energy dewatering.
(7) In one example for one city in Indiana, the sludge averages 15% solids (85% moisture or water). Using a prototype of the invention disclosed herein, the system was able to achieve 30% solids (70% moisture) at less than $0.01 per gallon. Mechanical dewatering also reduces the volume and weight of material to be transported by about half, thereby providing additional cost savings during transportation.
(8) This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter. Further embodiments, forms, objects, features, advantages, aspects, and benefits shall become apparent from the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows a perspective view of the claimed invention combined with a double drum drying system.
(2) FIG. **2**A shows a perspective view of the screen.
(3) FIG. **2**B shows a perspective view of a pump.
(4) FIG. **3** shows a perspective view of the mechanical dewatering apparatus.
(5) FIG. **4** shows a complete mobile mechanical dewatering apparatus with a screw press option.
(6) FIG. **5**A shows a top view of a slitter box.
(7) FIG. **5**B shows a bottom view of a slitter box.
(8) FIG. **6** shows a perspective view of an air knife.
(9) FIG. **7** shows a perspective view of a rubber seal.
(10) FIG. **8**A shows a portion of a compression plate without a seal.
(11) FIG. **8**B shows a cross section of the recess in the compression plate that retains the seal.
(12) FIG. **9** shows a compression plate with a rubber seal installed.
(13) FIG. **10**A shows a side view of a wedge wire.
(14) FIG. **10**B shows a perspective view of a wedge wire.
(15) FIG. **11** shows a perspective view of a power unit.
(16) FIG. **12** shows a polymer addition apparatus.
(17) FIG. **13** shows hydraulic cylinders attached to compression plates.
(18) FIG. **14** shows a perspective view of the top compression plate.
(19) FIG. **15** shows a compression plate comprising a conduit to introduce liquid material.
(20) FIG. **16**A shows a cross section of a disc from the slitter.
(21) FIG. **16**B shows a perspective view of the shaft that holds the discs.
(22) FIG. **16**C shows a cross section of a locking nut.
DESCRIPTION AND ILLUSTRATIVE EMBODIMENTS
(23) The following detailed description provides contemplated modes of carrying out embodiments of the invention. The description is not limiting, rather it illustrates the general principles of the inventions.
(24) Most biosolids from wastewater treatment plants comprise a liquid similar in consistency to skim or whole milk. The concentration typically ranges from 1-8% solids (92-99% moisture). Wastewater that is received in a typical wastewater plant will generally consist of a water content and smaller solids content. Typically, 99% of what comes into a wastewater plant is water. This water is usually treated biologically and discharged into a nearby stream or water body. The remaining 1% that consists of solids is separated and treated independently. The remaining solids are dewatered to 15-25% solids meaning the material still has 75-85% moisture in it.

(25) The inventions combine the use of pre-screening, chemical flocculation dewatering, non-chemical dewatering, conditioning, and/or leveling with high pressure compression and filtration to remove water and thereby reduce the volume of material. Material may be screened and dewatered with a centrifuge, belt press, screw press, or similar devices. Optionally, a polymer may be added to the material before dewatering to flock the liquid. The dewatering process thickens the material to approximately 15-30% solids (70-85% moisture) by weight. After this first thickening step, the material can be conveyed into the disclosed system.
(26) The disclosed inventions process the material that has about 75-85% moisture and reduces the moisture content to 55-75%. This dewatering substantially reduces the transportation and disposal cost. When this technology is used in conjunction with other thermal drying processes, it improves the economics and increases throughput by removing additional moisture that does not need to be removed thermally. By reducing the amount of water in the material, thermal dewatering may be avoided or minimized that reduces the effective cost of dewatering the material.
(27) Referring to FIG. **1**, the mechanical dewatering apparatus **10** can be combined with existing systems such as a double drum dryer system **20**. In one embodiment, the mechanical dewatering apparatus is operably positioned before the double drum dryer.
(28) Referring to FIGS. **2**A and **2**B, the process starts by using a pump **50** to convey the material **60** (not shown) through a screen **70**. The screen **70** separates from material **60** solids **80** (not shown) larger than about 2-10 mm or any other desired size. The screen **70** is preferably pressurized to move the material **60** through the screen **70**. This screening process ensures that no large solids **80** enter the compression chambers **90** that could damage the conveyor belt **100**. The solids **80** may be discharged and transported to a landfill. The screened material **60** is then pumped to a centrifuge **110** (FIG. **4**), belt press **120** (not shown), or screw press **130** (not shown) as known to a person skilled in the art
(29) During the pumping process, one or more polymers **140** (not shown) may be injected into the material **60**. The polymer **140** causes the material **60** to “flock”. Flocking aids in the separation of the solids **80** from the water in the material **60**. In the case of a centrifuge **110**, the specific gravity is magnified once the material **60** enters the centrifuge **110** which causes some water to separate from the material **60**, and the material **60** exits the centrifuge **110** at about 20-30% solids (70-80% moisture). The centrate **150** (i.e., water extracted from the centrifuge **110**), is discharged back into the wastewater plant. The material **60** may then be deposited into the slitter hopper **160** for further dewatering. In one embodiment, the mechanical dewatering apparatus **10** comprises two conveyor belts **100** that operate in parallel. One skilled in the art will realize that any combination of one or more belts **100** may be used without deviating from the claimed inventions.
(30) Referring to FIGS. **3** and **4**, the mechanical dewatering apparatus **10** has a slitter **170**, a conveyor belt **100**, one or more compression plates **180**. Material **60** is delivered to mechanical dewatering apparatus **10** through slitter **170**. Slitter **170** delivers material to conveyor belt **100** at predetermined volumes and velocities. The conveyor belt **100** transports material **60** from the slitter **170** and positions material **60** under compression plates **180**. Once the material **60** is positioned under plates **180**, the conveyor belt **100** stops, and the compression cycle begins.
(31) FIG. **4** shows a mobile mechanical dewatering apparatus **10** that may be moved to different water filtration plants. FIG. **5**A shows a top view of slitter **170** into which material **60** is deposited. FIG. **5**B shows a bottom view of slitter **170** from which material **60** exits the slitter **170** to be deposited on to conveyor belt **100**.
(32) Hydraulic cylinders **190** are operably connected with compression plates **180** to articulate plates **180** toward and away from belt **100**. The hydraulic cylinders **190** activate and move the compression plates **180** toward the conveyor belt **100** and onto the material **60** positioned on the conveyor belt **100**. In a preferred embodiment, mechanical dewatering apparatus **10** comprises sets of five compression plates **180** serially positioned above each of two conveyor belts **100**. Any number of compression plates **180** may be used without departing from the scope of inventions

disclosed and claimed.

(33) Each compression plate **180** has a seal **200** (FIGS. **7**-**9**) operably attached. The seal **200** may comprise one integral piece or up to four separate pieces forming the four edges of the seal **200**. When the cylinders **190** articulate compression plates **180** onto the conveyor belt **100**, the rubber seal **200** first contacts the conveying filter belt **100** to retain the material **60** under the compression plates **180** during compression without the material **60** escaping the compression chamber **90**. Pressure can be adjusted based on the size or composition of the compression plates **180**. In one embodiment, the compression plates **180** are about 30″×30″ and about 1″ inch thick. The compression plates **180** may be formed of any suitable metal, such as carbon steel, stainless steel, or other alloy. In one embodiment, the compression plates **180** are plated with a material that retards the adhesion of material **60** to compression plates **180**. In one embodiment, the compression plates **180** are at least partially chrome plated. As the compression plates **180** move toward belt **100**, rubber seal **200** compresses between the compression plates **180** and the belt **100**. The rubber seal **200** compresses proportionally at substantially the same rate as the compression plate **180** while maintaining a substantially intact seal. This eliminates or reduces the possibility that the material **60** will extrude out of the compression zone **90** between the compression plate **180** and the conveyor belt **100**.

(34) Referring to FIG. **8**B, the compression plates **180** have a seal extrusion **210** that removably engages the seal **200**. In one embodiment, the seal **200** is press fit into extrusion **210**. Extrusion **210** receives seal **200** into the compression plate **180**. Extrusion **210** receives the seal **200** inward as the compression plates **180** engages material **60**. Seal **200** substantially compresses into extrusion **210** to allow for substantially full compression of the plates **180** against the material **60** without any restriction created by the seal **200**. This novel feature allows for substantially higher pressure to be exerted onto the material **60** than allowed by systems known in the art.

(35) Conveyor belt **100** comprises a woven cloth **220** (not shown). Conveyor belt **100** carries the material **60** into position under the compression plates **180** and is supported by a wedge wire **230**. The hydraulic cylinders **190** cause the compression plates **180** to impart substantial force onto the material **60** and the conveying belt **100**. The wedge wire **230** is exposed to substantial pressure to support the impact and force being applied. Wedge wire **230** is preferably composed of carbon or stainless steel. In a preferred embodiment, the wedge wire **230** has a width of about ¼ inch and a height of about 5/16 inch. Positioned underneath wedge wire **230** are secondary and tertiary support structures, respectively **240**, **250** that bear the weight of the compression until the final force is applied to a floor **260**. Secondary and tertiary support structures **240**, **250** are preferably comprised of carbon or stainless steel. Floor **260** may be concrete or other suitable composite. The secondary and tertiary support structures **240**, **250** may comprise one or more drain pans **270** (not shown) to capture water that has been compressed out of the material **60** through the filter belt **100**. The drain pans **270** connect to one or more pipes **280** (not shown) that allow the removed water to flow out of the system **10** for disposal or reuse.

(36) A Programmable Logic Control (PLC) **290** (not shown) controls the compression rate and force. The PLC **290** allows the compression rate and force of the compression plates **180** to be programmed as desired depending on the material **60** being compressed. Multiple operating parameters may be programmed, including without limitation the amount of force exerted on the material **60**, the frequency of compression cycles, speed of compression plates **180**, speed of belt **100**, and others that may be known to a person of ordinary skill in the art.

(37) The cycle time is operator dependent and can be as little as 45 seconds or as long as the operator desires depending on throughput and desired dryness. For example, the longer the cycle time, e.g., the time that the compression plates **180** engage the material **60**, more water will be removed, the post-compression material **60** will contain less water, but the volume of material **60** processed over time will be reduced. Preferably, the cycle time for sludge is in the range of about 45 seconds to about 90 seconds.

(38) Once the programmed cycle completes, the hydraulic actuators **190** move compression plates **180** away from the material **60** and the conveyor belt **100**. The belt **100** may then rotate forward to convey the material **60** out of the compression chamber **90**. At or about the same time, one or more knives **300** (FIG. **6**) move across the faces **310** of the compression plates **180** removing residual material **60** that may remain on the faces **310**. Blades **300** may comprise an air knife or metallic knife. In a preferred embodiment, the air knife provides air pressure of at least 100 psi. The blades **300** may be mounted to a ball screw **330** or similar movement screws proximate the compression plates **180**.

(39) The material **60** is then conveyed off belts **100** and deposited into a screw auger **340** or second belt conveyor (not shown) for transport to a truck and/or holding chamber **350** (not shown). After the material **60** is removed from conveyor belt **100**, the conveyor belt **100** continues back to the slitter **170** to receive new material **60** to repeat the compression process. The material **60** processed as described above can be reduced in volume by as much as 50-75% depending on the original moisture content and material **60**.

(40) Referring to FIGS. **5**A and **5**B, the slitter hopper **160** comprises a cavity **360** to contain material **60**. Slitter **170** contains a plurality of individual discs **370** positioned in cavity **360** so that discs **370** engage material **60** as it passes from slitter **170** to belt **100**. In one embodiment, the discs **370** are stacked horizontally relative to the bottom of the slitter **170**. The discs **370** are preferably alternated with different diameters depending on the size that the operator desires. In one embodiment, the discs **370** are substantially round with respective diameters of about 4″ and 5″. The diameter size difference produces a ½″×½″ strip of material **60** that exits the slitter **170**.

(41) The discs **370** may have apertures **380** positioned and sized to accept shaft **390**. In one embodiment, shaft **390** has a substantially hexagonal cross section. The shaft **390** has a first end **400** and a second end **410**. The shaft **390** may include threaded portions **420** proximate first and second ends **400**, **410**. The threaded portions **420** receive a locking nut **430** to be positioned to hold the discs **370** to shaft **390** with a desired distance separating individual discs **370**. If the operator desires larger or smaller material **60** strips, the discs **370** may be sized accordingly and the distance between individual discs **370** may be adjusted accordingly as would be recognized by a person of ordinary skill in the art.

(42) The slitter **170** may be controlled by a variable frequency drive (VFD) (not shown) as known in the art. The VFD allows the operator to adjust the rate and/or volume of material **60** discharged out of the slitter **170** to the speed of the belt **100** on to which the material **60** is discharged. The width of the slitter **170** can be varied to the width of the belts **100**. The width of the slitter **170** may be adjusted by adding or removing discs **370**. Once the material **60** exits the slitter **170**, a self-leveling adjustable plate **450** (not shown) positioned proximate belt **100** substantially evenly distributes material **60** across the belt **100**. A belt control **460** (not shown) may be used to set and alter the movement of the belt **100** to convey the material **60** from the slitter **170** to the compression chamber **90** underneath the compression plates **180**. The belt control **460** may be manually adjusted or controlled by a VFD as would be known to one skilled in the art.

(43) The slitter **170** can be programmed to start and stop as needed to control the volume of material **60** discharged so as to avoid positioning material **60** on the belt **100** corresponding to areas between the compression plates **180** where no compression occurs. These systems can consist of a single, dual, triple, or multiple lines of compression plates **180**. In the case of a plurality of lines, a divider **470** (not shown) such as a V blade, may be used to deflect material **60** to a desired belt **100**.

(44) Referring to FIGS. **12** and **13**, an external hydraulic pump and oil reservoir **440** or an air over hydraulics system can be used to generate the pressure to articulate the compression plates **180**. Pressure may be applied up to 15.37 bar (223 psi). An 8″ hydraulic cylinder that is powered by a 4,000 psi power unit will generate up to 15.37 bar (223 psi) onto a 30″×30″ plate. Larger or smaller hydraulic cylinders **190** can be used depending on the desired pressure.

(45) The seals **200** may be comprised of rubber or other suitable compositions. The compression

plates **170** have recesses **500**. In preferred embodiments, the recess **500** may be either about ½″ or about ¾″ wide by ½″ or ¾″ deep. The recesses **500** extend continuously proximate the four outside edges **510**, **520**, **530**, **540** of the compression plates **180**. In one embodiment, the seal **200** comprises Urethane material. The seal **200** has a durometer range from about 40 to about 45. The depth of the recesses **500** on the compression plates **170** allow the seals **200** to retract during the compression cycle. The seals **200** absorb shock and compress into the compression plates **180** while still maintaining a seal around the material **60**. As shown in FIG. **8***b*, the recess **500** has a varying internal width allow seal **200** to expand and be retained by press fit into recess **500**.
(46) The seals **200** may comprise a durometer that is strong enough to handle the pressure forces created when the plates **170** compress against the conveyor belt **100** but yet flexible enough to compress inward and recess back up into the extrusion **210**. The seals **200** preferably have a memory characteristic to allow them to return to original shape once the compression cycle is finished.
(47) The compression rate and force are adjustable based on parameters set by the operator. The compression cycle may be held for any desired duration and at any desired pressure up to a maximum of about 223 psi. In one embodiment, the compression cycle starts at about 1 psi and increases to about 223 psi over about 60 seconds. A person skilled in the art will appreciate that the amount of force and duration of compression can be varied depending on the material being compressed, the desired water content, the desired volume of throughput or any combination thereof.
(48) Once the compression cycle is complete, the hydraulic cylinders **190** articulate the compression plates **180** away from the conveyor belt **100**. In one embodiment, the retraction rate is about 0.25 to 2.0 seconds per inch depending on the operator's throughput variables. The conveyor belt **100** may start to convey forward while moving blades **300**, with a minimum air pressure applied against compression plates **180** of about 80 psi, counter to the direction of travel of the belt **100**. Alternatively, the blades **300** may move in the same direction as the belt **100** to remove material **60** that may have stuck on the compression plate **180**. Alternatively, a pulsating air knife delivery system can be used to reduce the volume of compressed air. The belt **100** continues rotating around and begins the cycle over again at the slitter **170**. Depending on the material **60**, a high pressure water pump **550** (not shown) and spray bar **560** (not shown) may be used to wash the material **60** off of the belt **100**.
(49) During the compression cycle, a hopper **570** positioned above the slitter **170** stores material **60**. Once a compression cycle is complete, the hopper **570** deposits material **60** through the slitter **170** onto belt **100** and the compression cycle repeats. This allows the dewatering device (centrifuge, belt press, screw press, etc.) to operate continuously as desired by the volume of material **60** to be dewatered.
(50) An additional feature of this invention is the ability to provide a one-step dewatering and compression cycle instead of using an optional centrifuge, belt press, screw press, or the like. With this option, the operator may optionally add polymer **580** (not shown) to material **60**. If polymer **580** is used, the material **60** throughput will be higher. With this option, the operator may bypass the slitter **170** and pump material **60** into the inlet pipe **590** (FIG. **15**) on top of each compression plate **180**. The compression plates **180** move so that seals **200** engage belt **100** to form a substantially water tight seal. Liquid material **60** is then pumped into the compression zone **90** between the compression plates **180** and belt **100**. The pump **50** injects material **60** and creates the pressure needed to force the water through the belt **100**. Pressures up to 223 psi can be used.
(51) Once the compression zone **90** is substantially filled with material **60**, the pump **50** is turned off and the compression cycle begins. The compression plates **180** engage material **60** and belt **100** at a predetermined speed to extrude water from material **60**. A check valve (not shown) prevents material **60** from backfilling inlet pipe **590**. The compression cycles continue based on the predetermined cycle times. When finished, the compression plates **180** retract upward and the belt

100 unloading cycle commences as described above and the cycle repeats. The pressure exerted on the material **60** forces water out of the material **60**, through the conveyor belt **100**, and into drain pans **270**.

(52) The compression plates **180** may exert up to about 223 psi on material **60**. A belt press or screw press may exert up to about 10-15 psi. Support steel capable of supporting approximately, 200,000 lbs. of pressure is required to keep the deflection rates to a minimum for each 30″×30″ compression plate.

(53) The disclosed inventions my work in conjunction with devices like a double drum dryer or other thermal drying technologies. The disclosed inventions may be adapted to work in conjunction with existing dewatering technologies like a centrifuge, belt press, screw press, and the like. The disclosed inventions may include a screw press **300** positioned above the slitter machine **170** to first dewater the material **60**, move the material **60** to the slitter machine **170**, and expose the material **60** in the high pressure press **10** making the entire operation a complete replacement for the traditional dewatering machines.

(54) In another embodiment, the material may comprise of animal feed like a product such as Okra. The Okra may be about 10% solids (90% water). Okra may be added to cattle and hog feed for nutrient value. Byproducts such as Okra may be processed using the inventions disclosed herein to arrive at about 30% solids (70% water). Thus, one may dry the byproduct without adding heat that would adversely impact the nutrient value to be added to the cattle feed. Dewatering reduces weight and volume of the material and thereby reduces transportation expenses by about 70%.

Claims

1. A method for dewatering material comprising the steps of: a. delivering a material to a first storage; b. transferring the material from the first storage to a slitter, wherein the slitter separates the material into a plurality of clumps; c. depositing the plurality of clumps of material from the slitter to a conveyor belt, wherein the conveyor belt is porous to allow water to pass through without allowing solids to pass through; d. transporting the plurality of clumps of material on the conveyor belt from the slitter to at least one compression zone, the at least one compression zone comprising at least one high pressure press, the at least one high pressure press comprising at least one hydraulic actuator operably connected to at least one compression plate, wherein the at least one compression plate has a top surface, a bottom surface, and a plurality of side surfaces; e. compressing the material by at least one hydraulic compressor to articulate the at least one compression plate toward the material on the conveyor belt and applying a force to compress the material and holding the compression plates against the material and conveyor belt for at least 45 seconds; f. retracting the compression plates away from the conveyor belt; g. moving the conveyor belt to convey the compressed material away from the at least one compression zone; h. applying at least one knife to remove the compressed material from the at least one compression plate after a compression cycle; and i. directing water removed from the material away from the conveyor belt using at least one drain.
2. The method of claim 1 wherein the at least one compression plate comprises a substantially square plate having sides about 30 inches long and a thickness of about 1 inch.
3. The method of claim 1 wherein the at least compression plate comprises an alloy.
4. The method of claim 1 wherein the at least one compression plate further comprises an extrusion that engages a seal, wherein the extrusion receives the seal into the at least one compression plate, and the seal substantially compresses into a recess to allow for substantially full compression of the at least one compression plate against the material without any restriction created by the seal.
5. The method of claim 4 wherein the seal comprises an elastic material.
6. The method of claim 1 further comprising the step of supporting the conveyor belt with a wedge wire.

7. The method of claim 1 further comprising a step of controlling the rate of activation of the at least one hydraulic compressor and the force applied by the at least one hydraulic compressor using a programmable logic control.
8. The method of claim 1 further comprising separating the material into a plurality of strips as it exits the slitter using a plurality of discs and then depositing the plurality of strips of material at a substantially even height on the conveyor belt.
9. The method of claim 1 further comprising controlling the volume of material deposited from the slitter to the conveyor belt and the speed of the conveyor belt using a variable frequency drive.
10. The method of claim 9 wherein a programmable logic control selectively activates the variable frequency drive.
11. The method of claim 1 wherein the at least one compression plate exerts a force of less than about 223 pounds per square inch on the material.
12. The method of claim 1 further comprising a step of introducing a polymer to the material before it enters the slitter wherein the polymer causes the material to flock thereby reducing water content from the material before it enters the slitter.
13. The method of claim 1 wherein the material is compressed for about 60 seconds during each compression cycle.
14. The method of claim 1 further comprising a step of distributing the material at substantially even depth across conveyor belt.